Amitesh Kumar

Personal Information

Name: Amitesh Kumar
Total Years of Experience: 15+ Years
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Executive Summary

Overall summary:

15 Years of Development experience in Linux Embedded, Network and Cryptography..

Detailed summary:

- **Expertise in below domains**
  1. **Network programming/Protocol implementation**. Used c language to implement HTTP GET, POST(partial), CGI, SMTP, DNS, OMCI packet analyser, multicast IPTV (MPEG2TS) stream analyser, etc. Did more than 11 router board bring-up work and developed almost all the router components. Did a few PoCs in DPDK.
  2. **Cryptography in c and rust** : Used OpenSSL, DTLS, PKI, X509, ACME, LibSodium, Nettle, Botan, Noise protocol, KMIP, TPM, HSM, libcrux, openMLS, RustCrypto etc
  3. **Extensive DNS development experience in client, proxy, firewall**
  4. **Wifi Router OpenWRT based product development**
  5. **Rust expertise on below domains**.
     a. **Rust for Embedded System (Embedded linux and bare-metal)**
     b. **Rust for Network TCP/IP programming**
     c. **Rust for WebAssembly for Edge/IoT devices**
     d. **Rust for http client and server implementation**
     e. **Rust for AI/ML inference and pipelining**
     f. **Rust for Post Quantum Cryptography**
- **Languages Expertise :** Highly expert in **C, Lua, Go, Rust** languages and working in all these languages on a full time basis. having experience with other languages as well like Kotlin, scala, php, pearl, awk, shell scripting, python etc
- **Decent Expertise** in below domains
  1. **Embedded System Board bring-up and cross compilations**
  2. **AI (Machine Learning)**
  3. **Streaming domain**
  4. **Server side development using C, Go**
  5. **Client side development using C, Rust**
  6. **IoT / Cloud based development using WebAssembly / WASMEdge in Go and Rust**
  7. **Payment Industry (card reader)**
  8. **Automotive domain work at RTE layer**
  9. **XDP (eBPF) based network datapath experience.** Good understanding of **linux kernel datapath** (TCP/IP stack), used hooks of datapath for development firewall, load balancer.
  10. **UEFI, Bootloader, payload, GRUB based work** for customizing initrd/rootfs and securing it using TPM, LUKS, Clevis, PKI signing, PBKDF, PKCS #5, PKCS #11

Project Summary

SN	Project Name	Role/Competency	Tools (Platforms/OS to be included)
41	GUI, CLI, CMD developments	Development	nushell, slint, dioxus, CobraCLI
40	Noise protocol implementation	Development	c, go, rust, AEAD, TLS, NewHope
39	Post Quantum Cryptography in C and Rust	Development	c, rust, libcrux,MLS, KEM, HPKE
38	DNSSec, DoT, DoH, DoQ.implemention	Development	Rust, hickory, DNS
37	NLP and pipelining using rust-bert and ONNX	Deployment	Rust, ONNX, Ort, Rust-bert, AI, ML, inference, pipeline
36	WASMEdge work in C, Rust	Development	WASM, WASMEdge, rust, c
35	XDP (eBPF) work, Kernel Side in c and User side in Rust and Go	Ramp up, Development, PoC	XDP, data path, tcp/ip, C, go, rust, dissector, hash-maps
34	Bootloader GRUB scripting for rootfs customization and security using cryptsetup, pbkdf, LUKS, clevis, TPM, systemd-cryptenroll etc	Scripting, Configuration, Cryptography, Integration	Coreboot, UEFI, EFI, BIOS, Payload, LUKS, PKI, Encryption, GPT, MBR, Payload, GRUB2, clevis, TPM, cryptsetup, go-rootfs, dracut, initrd, initramfs

33	AI inference in rust lang	Development	rust, candle, burn, tensorflow-lite
32	Client Development in rust lang	Development	rust, tokio, async, reqwest, hyper, axum
31	Server Development in go, gin gonic	Development	golang, gorilla, gin gonic, autotls, caddy, nginx
30	Stream analysis using ffmpeg and vlc	Freelancing, Development	ffmpeg, codec, filters, rust, lua, vlc
29	Knot resolver lua scripting for DNS ACL firewall and DNS intranet implementation	Freelancing, Development	DNS resolver, relay, proxy, ACL, Knot resolver, lua, luajit
28	HTTPs and data encryption using zero key setup and in-app DNS resolution	Freelancing, Development	Linux, c, libsodium, certbot, mTLS,ACME, libcurl
27	vending machine (MDB) and Credit card reader (POS/EMV) in c	Freelancing, Development	EMV cardreader, Linux, c, DUKPT, libcurl, ISO 7816, 1443, MDB protocol
26	Automotove Cloud Client	Development, Debugging, PoC	C, Linux, autosar, ECU, OpenAuth, Json, c-curl
25	ARM cortex TrustZone, mbed based bootloader	Debugging, Release	c, cryptography, ARM PSA, ARM TZ, TEE, TPM, baremetal C, RISC-V
24	OpenWRT	Development	Linux, c, tr069, OpenWRT
23	Firewall load balancer	PoC, Development, Requirement gathering	Linux, Kernel 5.4, bridge driver, teamd driver, bonding driver, DPDK, nfconntrack
22	MPEG stream analysis using libpcap for jio IPTV(current project)	Development	C, MPEG TS, libpcap, multicast socket api, IPTV
21	PoC of audio sample analysis using machine learning librosa and keras	Development	Keras,Tensor Flow, python, librosa
20	Computer Vision Mask R-CNN implementation in python keras for JioTV automation in custom channel logo dataset	Development	Keras,Tensor Flow, Mask R-CNN model,
19	OMCI Analyser development in python	Development	ITU-984.4 standard, python
18	TeraTerm automation to run some test case on enterprise switch	Development	Windows, TeraTerm scripting, Linux, telnet, nc, python
17	JioTV/Android TV automation	Development	JioTV (Android based STB), python
16	Android app development for Bluetooth based multi-slave test execution	Ramp up, Development	Android, Kotlin, Bluetooth, RFComm sockets
15	STB/TV big data analytics	Ramp up, Debugging	Perl, Scala, Apache Spark, Hadoop
14	WiFi Mesh PoC	PoC	OpenWRT, batman-adv, wifi, mesh
13	NFV deployment using OVS base OpenWRT router and OpenNFV	Build, compilation, PoC	OpenWRT/LEDE, OpenVSwitch, OPNFV, OpenStack, compass4nfv
12	IoT (esp8266) energy meter and Wifi mesh networking	Development, PoC	Arduino C, ESP8266 API
11	CPE (OpenWRT based Project)	Developement	C, OpenWRT, OpenVSwitch, OpenFlow, OpenStack, neutron, python, c
10	OctoConnect (OpenWRT based project)	Development	C, OpenWRT
9	KeySecure (Crytography)	Development, Automation	C, C++, CentOS, OpenSSL
8	GEPON (Passive optical based Network)	Development, Debugging and Documentation	Montavista Linux, Power PC
7	IoT, Phase 1 (Internet of Things)	Ramp up, Development	Embedded Linux, ARM,
6	Debugging of various components like SNMP, DHCP, RIP, VPN, etc	Debugging	C, lua, syslog, printk, kgdb, wireshark, tcpdump, linux

	according to bug report.		
5	FVS318Gv2 kernel, userspace porting, component integration, customization and debugging	Build, customization and debugging	Gmake, arm toolchain, c, bash script, autotools, linux
4	IPv6 integration and implementation	Development and debugging	Socket library, DHCPv6, RADVD, c, linux
3	CLI (Command Line Interface)	Development	Lua, xml, clish, sqlite, c++, linux
2	OpenCV based monitoring (CDAC Project)	Development	C, OpenCV, Linux
1	SMS Database in Symbian (Engg. Project)	Development	PyS60, python, symbian

Skills

Operating System	Embedded Linux, Linux, Symbian, Montavista, OpenWRT, Android
Programming languages	C, Lua, Python, Go, Rust
Scripting Languages	Shell Scripting, Tera Term scripting, Perl
Domains	IT, Telecom, TCP/IP Networking, Cryptography, Machine Learning, Streaming, Automotive,
Hardware	ARM, MIPS , RISV-V and PPC based boards, router boards
Tools	GNU toolchain for ARM and MIPS, Makefile, vim, ctags, autotool, buildroot, wireshark, tcpdump, QEMU, OpenWRT, Tensorflow, Keras, STB, ffmpeg, gstreamer, libpcap, libcurl, ACME, x509, PKCS 11, grub, LUKS, clevis, coreboot, nginx, caddy, tokio, zig

Employment History

Employer Name	Designation	Duration	Reason of Job change
Aeris Communications, Bangalore	SSE	14 Aug 2023 to 31 March 2024	Recession/Lay-off
Infobell IT, Bangalore	PMTS	9 Dec 2022 to 20 July 2023	Recession/Lay-off
Radisys, Bangalore	Staff Engineer	23 May 2022 to 3 Nov 2022	Recession/Lay-off
Infinity Labs, Noida	Architect	21 Nov 2021 to 29 April 2022	Presales work instead of Development
Reliance Jio Infocomm, Navi Mumbai	Device Test Manager	22 Nov 2018 to 17 Nov 2021	Career Growth
Xavient Software Solutions, Noida	SSE	16 Jan 2017 to 20 Nov 2018	Career Growth
Nimbuzz, Gurgaon	Firmware Engineer	16 June 2016 to 13 Jan 2017	Company shutdown
SafeNet (Gemalto), Noida	SSE	15 Feb 2016 to 15 June 2016	Support work instead of Development
Sasken Communication, Hyderabad	Senior Engineer	9 Feb 2015 to 12 Feb 2016	Career Growth
TeamF1 Networks (subsidiary of D-Link), Hyderabad	Software Engineer	7 Mar 2011 to 5 Feb 2015	Career Growth

Education Details

College/University	Degree	%	Specialization	Duration
Gautam Budh High School, Jehanabad	10th	76	PCM	2001 to 2002
A N College, Patna	12th	62	PCM	2002 to 2004
Apeejay College of Engineering, Gurgaon	BE	67	Electronics and Comm.	2006 to 2010

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CDAC, Hyderabad	Post Graduate Diploma	60	Embedded System Design	2010 to 2011

Project Details

41.GUI, CLI, CMD developments

Project Description: Multiple project where User Interface was developed in Cmd, Webview, GUI
Tools Libraries used : Slint, fyne, nushell, dioxus, rust, go
Responsibilities: Developed fully functional, not so beautiful user interface with auto completion CMDs, cross platform app for windows, linux, mac, iOS, android. Wrote cross platform shell scripting in nushell.
Duration and location:3 month, Remote

40. Noise protocol implementation in c, go, rust

Project Description: Noise protocol framework is for security like SSL/TLS in distributed environment like Mesh or P2P connection
Tools Libraries used :
Responsibilities: Used noise-c, snow, newHope for implementing noise and creating a secure socket in c, rust and go lang.
Duration and location:1 month, Remote

39. Project Name: Post Quantum Cryptography in C and Rust

Project Description: Quantum computers are able to break all cryptographic algorithm and because of it PQC Post quantum cryptography is implemented in many projects in c and rust language
Tools Libraries used : Libcrux, OpenMLS, KEM, HPKE, RustCrypto, KEM-TLS
Responsibilities: Wrote new code or refactored existing, especially to key exchange and key derivation (KDF) for quantum safty. Working on MLS for secure messaging end to end encryption.
Duration and location: WIP , Remote

38. Project Name: DNS client and resolver implementation using hickory rust crate

Project Description: In-App implementation of async DNS client and DNS resolver/proxy
Responsibilities: using hickory library for client and resolver, implemented modern DNS like DNSSec, DoT, DoH, DoQ
Duration and location: 1 Months, Remote

37. Project Name: NLP and pipelining using rust-bert and ONXX runtime ort

Project Description: Inference and pipelining implementation using rust-bert crate
Responsibilities: Used huggingface pret-rained model for multiple operations like Translation, Summarization, Question-Answering, Keywords extraction.
Duration and location: 2 Months, Remote

36. Project Name: WebAssembly WASMEdge work in Rust

Project Description: WebAssembly is common runtime for embedded system, mobile, desktop, cloud and because of it porting multiple existing components especially client and server in WASM.
Responsibilities: Developing using mostly rust language using async frameworks like tokio, reqwest, axum, hyper, Rust-Crypto, rustls, webpki etc.
Duration and location: 1 Month, Remote

35. Project Name: XDP (eBPF)

Project Description: Packet datapath early hooks in XDP development
Responsibilities: Implementing monitoring, firewall, load-balancer in XDP. As XDP itself is new framework, learning it and implementing it after PoC.
Duration and location: 2 month, Remote

34. Project Name: GRUB scripting for rootfs customization and security

Project Description: Customization and securing booting process
Responsibilities: Wrote custom grub script that mount encrypted partitions, mount, read custom config stored in that partition then based to that selecting a kernel, rootfs. Modifying rootfs/initrd using dracut and gobusybox.
Tools, OS and Languages: Coreboot, UEFI, EFI, BIOS, Payload, LUKS, PKI, Encryption, GPT, MBR, Payload, GRUB2, clevis, TPM, cryptsetup, go-rootfs, dracut, initrd, initramfs
Duration and location: 2 Months, remote

33. Project Name: AI inference in rust lang

Project Description: Using pre-trained models for doing inference
Responsibilities: Using rust framework like candle, burn and using pre-trained model of tensorflow, pytorch etc to perform some operation then creating a pipeline to verify output of one mode then after some processing giving it to another AI inference and so on.

Duration and location: WIP, Remote

32. Project Name: Client Development in rust lang
Project Description: HTTP client development in rust in different projects since 2018
Responsibilities: Used rust for writing async client using tokio, reqwest, axum, hyper, Rust-Crypto, rustls, webpki etc.
Duration and location: Combined 8 Years, Different projects

31. Project Name: Server Development in go
Project Description: HTTP client development in rust in different projects since 2015
Responsibilities: Used golang gin gonic, iris, gorilla etc framework for writing HTTP server, route and end-points.**Duration and location**: Combined 10 Years, Different projects

30. Project Name: Stream analysis and modification
Project Description: Stream analysis and modification using ffmpeg and vlc at codec, stream, mux, demux or packet level
Responsibilities: Using ffmpeg to split streams like separated audio, video and used inbuilt filters of ffmpeg to modify the stream then mixing different streams. used vlc-rust and vlc-lua library to automate steam playback. used gstreamer for pipeline based work.
Duration and location: 15 Days : Remote

29. Project Name: Knot resolver lua scripting for DNS ACL firewall and DNS intranet
Project Description: DNS ACL requirement for controlling local DNS proxy and providing hostname to all hosts in LAN
Responsibilities: wrote lua scripts for dynamically configuring the firewall rules that allow/deny user, modify dns response, redirect requests. wrote lua code that get response from another server to user domain name like *user1.company.in* having ip like *192.168.1.86.* in this way DNS was implemented at intranet level host all hosts.
Duration and location : 1 Months : Remote

28. Project Name: HTTPs and data encryption using zero key setup and in-app DNS resolution
Project Description: http client sending some device specific data line link status, ip, routes, config file. Requirement was to secure the data and make sure data is not tampered while in-route
Responsibilities: wrote client code in libcurl. Using libsodium encrypted the data, encryption header, checksum, token etc was sent in header for verification on server end. Used certbot for certificate management and client certificate mTLS. As the server team had problems understanding server side code, i wrote server side code as well in PHP.
Duration and location : 1 Months : Remote

27. Project Name: vending machine (MDB) and Credit card reader (POS/EMV) in c
Project Description: Developing vending machine and card reader software POS in c and displaying on LCD display using vendor specific API/framework
Responsibilities: implemented core payment logic of reading MSR, RFID tap, Chip and generating card data as per EMV standard, display on LCD, vending machine invoice text logic, static multi-language support using extended ASCII, DUKPT for token generation etc.
Duration and location : 6 Months : Remote
Team Size : 8+ members

26. Project Name: Automotive Cloud Client
Project Description: working on C, Linux, C-Curl, Json, ECU based firmware to handle status reporting and operations like door close/open/lock, temperature control and other car functions through cloud
Responsibilities: worked on debug and releases as per requirement.and bug report
Duration and location : 7 Months : Aeris Comm, Bangalore

25. Project Name: ARM cortex TrustZone, mbed based bootloader and RISC-V based bootloader
Project Description: Worked on CPU microcode based bootrom, payload based booting involves signed payloads, TPM as root of trust, ARM TrustZone
Responsibilities: Working on debug and releases as per requirement.and bug report
Duration and location : 6 Months : Infobell IT, Bangalore
Tools, OS and Languages: Embedded C, TPM, code signing, Public cryptography, ARM TrustZone, ARM PSA, mbed, applied cryptography
Team Size : 9

24. Project Name: OpwnWRT board bringup and TR069

Project Description: OpenWRT board with wifi 6E and BSP integration. TR069 handler work
Responsibilities: Doing build, debug and integration work for board bringup. Wrote TR069 handlers as per requirement.
Tools, OS and Languages: C, Linux, OpenWRT, TR069
Duration and location : 5 Months : Radisys, Bangalore

23. Project Name: Firewall load balancer requirement

Project Description: Firewall load balancer is supposed to balance network load in a persistent way across multiple firewalls
Responsibilities:

- Trying to acquire project. Dealing with technical aspect of product requirement.
- Working on requirements
- Doing PoC on kernel level bridge, teamd and bonding driver. Able to select a slave port based on dynamic criteria to send packet outside. Work is done in c, Linux kernel 5
- Used DPDK and wrote a sample application to demonstrate basic functionality. And did PoC for stateful connection using SFT tables.

Duration and location : 5 Months : Infinity Labs, Noida
Tools, OS and Languages: C, Linux kernel 5.14, teamd driver, bonding driver, bridge driver. DPDK

22. Project Name: MPEG stream analysis using libpcap for jio IPTV

Project Description:
JioTV automation work using Multicast IP packet analysis in c code
Responsibilities:
As manual video/audio quality measurement is not possible all the time, I wrote an application using c, multicast, libpcap to join multicast MPEG streams, capture using libpcap, typecasting to desired header and counting lost packets.
Duration and location : 3 Days : Reliance Jio, Navi Mumbai
Tools, OS and Languages: C, MPEG TS, libpcap, multicast socket api, IPTV

21. Project Name: PoC of audio sample analysis using machine learning librosa and keras

Project Description:
JioTV automation work using audio signal machine learning
Responsibilities:
using Keras and libroas, extracted features from audio sample and matched with channel logo sound dataset to detect channel being played.
Duration and location : 2 Weeks : Reliance Jio, Navi Mumbai
Tools, OS and Languages: Keras,Tensor Flow, python, librosa

20. Project Name: Computer Vision Mask R-CNN implementation in python keras for JioTV automation in custom channel logo dataset

Project Description: JioTV automation work using computer vision machine learning
Responsibilities: Did learning work of tensor flow on MNIST dataset
Created custom dataset of channel logo by capturing video using Inforce 6420 Board hdmi input and android camera1 API. Used OpenCV to capture frame of video and finally cropped only logo part of frame. In this way created channel logo dataset.
Used Mask R-CNN to find channel logo in frame grabbed. Similarly used U-NET to do similar work.
Duration and location : 2 months : Reliance Jio, Navi Mumbai
Tools, OS and Languages: Keras,Tensor Flow, Mask R-CNN model, Custom dataset of logos

19. Project Name: OMCI Analyser developpment in python

Project Description: Writing a OMCI analyser from scratch to analyse OMCI packet capture in hex. It was used for JioFibre testing.
Responsibilities: Studied ITU-984.4. Wrote a python module to take Hex input of OMCI message from file, ME Class information from another file. Dissected OMCI message as per Fixed header and ME Class information from config file. ME class info file was generated and some parts written by testers. Generated report for end user in python.
Duration and location : 2 months : Reliance Jio, Navi Mumbai
Tools, OS and Languages: ITU-984.4 standard, python

18. Project Name: TeraTerm automation to run some test case on enterprise switch

Project Description: Project to automate some common manual GEPON provisioning work
Responsibilities: Got commands from testing team that they execute for configuring GEPON. Generated a TeraTerm script file based on user configurable json file by python. Json file has some common info like username, password, IP address of switch, port id, vlan id, command list. Executed and analysed output by python and generated report file.
Duration and location : 2 months : Reliance Jio, Navi Mumbai
Tools, OS and Languages: Windows, TeraTerm scripting, Linux, telnet, nc, python

17. Project Name: JioTV/Android TV automation

Project Description: Project was automating manual operation of channel change and its resolution and response measurement
Responsibilities: Used adb to connect to JioTV set-top box
Using input commands to change the channel and doing many other operations that were done by remote. Automated all these things using a python script. Collected the data every every channel change and generated report
Duration and location : 2 months : Reliance Jio, Navi Mumbai
Tools, OS and Languages: JioTV (Android based STB), python

16. Project Name: Android app development for Bluetooth based multi-slave test execution

Project Description: A single master android app and multi-slave app connected over Bluetooth, where master will send test case to slaves and slaves will execute and return back return. Master was aggregating result from all slaves and displaying in master app. Master was also sending daily report mails.
Responsibilities: Rampup to android as completely new to it. Learnt kotlin language for app development. Learnt android app development.
Developed unified android app in kotlin where same app is used by client and server. Used json, Bluetooth RFComm sockets for rpc communication between master and slave. Use java-mail API to send mails.
Duration and location : 4 months : Reliance Jio, Navi Mumbai
Tools, OS and Languages: Android, Kotlin, Bluetooth, RFComm sockets, ffmpeg, gstreamer, box4

15. Project Name: STB/TV big data analytics in Apache Spark and Scala

Project Description: Project was big data analytics project where TV/STB data was analysed
Responsibilities: Rampup to big data and scala. Debugging of issues reported (internally or externally)
Duration and location : 3 months : Xavient Noida
Tools, OS and Languages: Perl, Scala, Hadoop, Spark

14. Project Name: WiFi Mesh PoC using batman-adv

Project Description: PoC of layer 2 based dynamic wifi mesh using open source batman-adv.
Responsibilities: Studied various mesh topology like L3 mesh openmesh, OLSR, batman/alfred
Found only one L2 mesh opensource solution i.e. batman-adv
Installed batman-adv in OpenWRT and finished the topology
Duration and location : 1 months : Xavient Noida
Tools, OS and Languages: OpenWRT, batman-adv

13. Project Name: NFV deployment using OVS base OpenWRT router and Opnfv

Project Description: NFV PoC for creating full test environment for CPE project
Responsibilities: using compass4nfv installed opnfv.
Installed openstack from devstack and explored horizon
Duration and location : 3 months : Xavient Noida
Tools, OS and Languages: OpenWRT/LEDE, OpenVSwitch, OPNFV, OpenStack
Team Size : 7

12. Project Name: IoT ESP8266

Project Description: Cloud enabled Energy meter (and other sensors) using ESP8266 32 Bit microcontroller and Hall effect based ACS712 sensor.
Responsibilities: Written code for running WiFi AP as well as client. Written code to run http server and handling client requests. Saving user input to EEPROM and reading those configs at the time of reboot. Written code to read the ACS712 energy meter and sending data to a server in json format.

Duration and location : 1 months : Xavient Noida
Tools, OS and Languages: Embedded Linux : C, Arduino IDE, ESP8266/NodeMCU

11. Project Name: SDN CPE

Project Description: CPE is a Cloud enabled OpenWRT based CPE that provides wifi solution using OpenVSwitch
Responsibilities:

- Worked on very recent and partially supported board IPQ40xx and IPQ806X. Fixed some basic firmware level problem.
- Rampup for OpenVSwitch, OpenFlow, ONOS
- Written daemon for switch port status detection, DNS probe detection, ping detection
- Enabled WDS (wireless distribution) in router.

Duration and location : 9 months : Xavient Noida
Tools, OS and Languages: Embedded Linux : C, OpenWRT, OpenVSwitch, OpenFlow
Team Size : 7

10. Project Name: OctoConnect

Project Description: OctoConnect is a OpenWRT based project that provides free wifi in public places.
Responsibilities:

- Added features in router LAN based extender where extenders are also based on OpenWRT
- Developed complete http based web interface (CGI) using c language and CGI 1.1 specifications. This included GET and POST method of HTTP. Written WAN setting management in c. http file upload for firmware upgrade in c. http file download for tcpdump capture download from browser.
- Enabled PPPoE in backend and wrote its CGI management for user configurations. Enabled WDS (wireless distribution system) in router.
- Enabled LAN port management so that individual port can be enabled or disabled
- Enabled a mesh networking where LAN and WAN was completely wireless using AP and STA wireless modes.
- Written proximity feature where logging user's wifi scan (probe) when user passes by and logging all user's details in server. Code change was done in hostapd probe function, where I logged user's MAC address whenever user's wifi device sends a probe.

Duration and location : 7 months : Nimbuzz Gurgaon
Tools, OS and Languages: Embedded Linux : C, OpenWRT, Radius
Team Size : 2

9. Project Name: KeySecure

Project Description:

- KeySecure is project for securing various network based data transaction like in bank every transaction is secured/ecnrypted with KeySecure.
- This includes Hardened CentOS with secure booting, code signing, secure sockets, CA signing, x509 certification, encryption key storage.
- Client side of the KeySecure includes various protocol like NAE(proprietary) and KMIP.

Responsibilities:

- Ramp-up to cryptography concepts, KeySecure components and protocols like NAE(Network attached encryption) and KMIP
- Automation of Coverity, Build machines etc using jenkin
- Scripting of build using perl.

Duration and location : 4 months : SafeNet/Gemalto Noida
Tools, OS and Languages: Xen Hypervisor, CentOS : KMIP, NAE protocol, OpenSSL, C/C++, perl
Team Size : 8

8. Project Name: GEPON

Project Description:

- Gigabit Ethernet Passive Optival Network is a technology that resides at ISP side and used for internet access.
- It is a combination of OLT (optical line terminal) and ONU (optical network unit)
- It is project where ARM based hardware was used in OLT. And project was creating application software for OLT.

Responsibilities:

- Ramp up for PON technologies.
- Working on management code in c for firmware upgrade and configuration management/reset.
- Working of management code in c for ONU authentication.
- Working on Datapath and bandwidth.

Duration and location : 8 months : NEC Noida
Tools, OS and Languages: QEMU, Beth : Montavista Linux : C
Team Size : 73

7. Project Name: Internet of Things

Project Description:

- Internet of Things is used by small hardware like mobile phones and similar hardware to access cloud through a SDN (software defined networking) controller.
- It was porting a customized kernel on Snapdragon ARM board. And implementing basic feature of IoT communication.

Responsibilities:

- Ramp up for IoT.
- Porting work which includes fixing compilation issues.
- Writing some management code to create some basic communication between the “thing” like wifi,bluetooth enabled devices and Wifi gateway. So that the “thing” can access cloud or internet.

Duration and location : 2+ Months : Sasken Hyderabad
Tools, OS and Languages: Makefile, vim : Embedded Linux : C
Team Size : 4

6. Project Name: Debugging of various component like SNMP, DHCP, RIP, VPN, WIFI, etc according to bug report.

Project Description:

- Fixing bugs reported by QA or customer or client.

Responsibilities: Reproducing issue, narrow downing bug, localisation in code, fixing and verification.
Duration and location : 3 years and 11 months (in parallel with other projects) : TeamF1 Networks, Hyderabad (Development centre of D-Link)
Tools, OS and Languages: syslog, printk, kgdb, wireshark, tcpdump : Linux (user space and kernel space) : C

5. Project Name: FVS318Gv2 kernel, user space porting, component integration, customization and debugging

Project Description:

1. It was a new router and work was to make it a router by making the bsp, kernel, driver, userspace daemon up and running.
2. It included basic BSP validation.
3. Applying patches of kernel and changes in vanilla kernel and fixing compilation error. Then doing basic sanity manually that that driver or kernel sub system is working or not, etc.
4. Integration/customization/debugging of user space components required by the customer.
5. Sub projects specific to FVS318Gv2 :
    1. **Firmware header validation**
    2. **External Authentication using RADIUS C API**

Responsibilities:

- I Took a reference project and modified and customized it according to the new hardware. This includes kernel porting and then component level customization and debugging.
- Dev acceptance test.
- **Firmware header validation :** written code in c to read firmware as ascii file and reading product ID in header and matching with device product ID and then only allowing flashing firmware.
- **External Authentication using RADIUS C API** : wrote c code to authenticate user with external RADIUS server.

Duration and location : 1+ years : TeamF1 Networks, Hyderabad (Development centre of D-Link)
Tools, OS and Languages: Gmake, arm toolchain, autotools : Linux : Bash Script, Make script

4. Project Name: IPv6 integration and implementation

Project Description:

- IPv6 was an enhancement request by customer and all routers had only IPv4. After this enhancement router became capable of IPv6 and IPv4.
- L5/Application layer requirement was integration of DHCPv6 server and RADVD for LAN. And DHCPSv6 for WAN as ISP connection type.
- L4/Socket layer work was add/upgrade existing code written only for AF_INET to AF_INET6. Some address and subnet validation was needed to make configuration user friendly.
- L3/Routing layer was mostly used in debugging and mostly theoretical analysis.
- Sub projects:
  - **nslookup implementation in C**
  - **SMTP client implementation in C**

- Learning IPv6 and implementing in linux kernel and userspace.
- DHCPv6 and RADVD debugging.
- upgrading IPv4 socket to Ipv4/IPv6 or writing similar IPv6 socket code.
- Analyzing functionality of these datagram in routing, multicasting and neighbour discovery.
- **nslookup implementation in C** : using getaddrinfo, getnameinfo functions, AF_UNSPEC and sockaddr_storage implemented nslookup as the busybox nslook had problem.

- **SMTP client implementation in C** : implemeted SMTP client with the help of socket API.

Duration and location : 2+ Years : TeamF1 Networks, Hyderabad (Development centre of D-Link)
Tools, OS and Languages : Socket library, DHCPv6, RADVD : Linux : C
Team Size : 6

3. Project Name: CLI (Command Line Interface)

Project Description:

- Command Line interface is needed to provide a telnet based console to user, so that user can get friendly commands options grouped (like wan, lan, firewall, etc).
- Work included to write commands like “net wan static ip address <input>” in XML file.
- Writing a business logic handler in Lua to validate and show error message and at last save the user input in permanent database, so on next reboot user will not have to configure it every time.
- Writing/modifying database handler and these handler written in C can interact with other process and with kernel to start a daemon/process and enable/disable some feature in kernel or insert modules.
- Designing and implementing business logic and creating/modifying SQL schema accordingly.
- Adding commands in group and structured way in XML to make the router configuration easy for user.
- Writing management code for business logic in lua
- Adding/modifying SQL table handler (C language functions) and with IPC mechanism like pipe, FIFO, exec, system, sockets, ioctl, sysctl configuring/starting/restarting daemons like http, snmp, dhcp, 802.11, route, etc.

Duration and location : 6 Months : TeamF1 Networks, Hyderabad (Development centre of D-Link)
Tools, OS and Language : Clish, sqlite : Linux : C, Lua, XML, C++
Team Size : 2

2. PGD Project Name: OpenCV based monitoring system

Project Description:

- It was Post Graduation project in which we (5 Engineer) worked to create a public surveillance with the help of web-cam and OpenCV.
- It was counting the total people entered by detecting the face in one frame all the time at gate of a public place like mall, hotel, etc.
- Learning OpenCV, exploring its API
- Implementing a people counter I.e. Person(s) has to be detected by face or another way (OpenCV API) and count has to be displayed on monitor..

Duration and location : 2 Months : CDAC Hyderabad
Tools, OS and Languages: OpenCV : Linux : C
Team Size : 4

1. BE Project Name: SMS Database in Symbian

Project Description:

- It was a final year project and it was working on application level writing some management code in python.
- It was creating a dbms system for SMS inbox. i.e. Create a database of SMS inbox then search and show according to keywords like phone number or person name or date or combination of these.

Responsibilities:

- Learning Symbian Framework, Python language, PyS60 API
- Writing code to access mobile's SMS and reading each SMS and creating a database. Take some user's input like name or phone no then generate a db query and execute using python and return the desired result with some GUI API.

Duration and location : 8 months : Apeejay College of Engg, Gurgaon

Tools, OS and Languages: PyS60 : Symbian : Python